Predicting Risk, Pricing the Future

Tweedie-optimized ML for profitable insurance pricing

39,928 TRAINING RECORDS

13,310 TEST RECORDS

88.9% ZERO CLAIM RATE

1 BUSINESS RISK

Mispricing today creates losses tomorrow.

- Underpricing high-risk customers drives systematic losses.
- Overpricing low-risk customers leads to churn and lost growth.
- Adverse selection spirals weaken portfolio quality.

2 PREDICTION FRAMEWORK

Data to Insights.
Leverage key policy, customer and behavioral signals.

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Predict Future Loss
Estimate expected loss and claim likelihood for each policy.

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Smarter Pricing
Use predictions to price fairly, competitively, and profitably.

3 MODEL VALUE (METRICS)

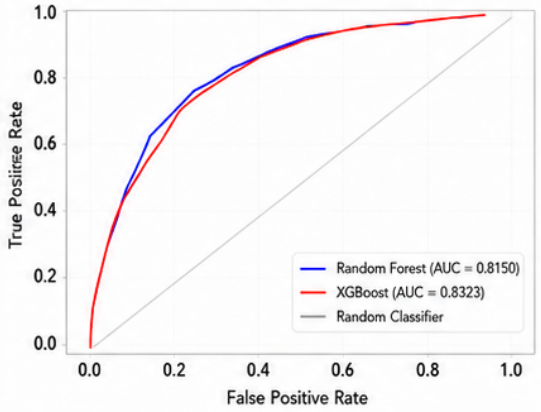
Strong performance where it matters most.

LOSS PREDICTION (LC & HALC)

Lower MSE = Better

XGBoost (Tweedie)	VS.	Random Forest
Lower MSE (Better)		Higher MSE

ROC-AUC CURVE COMPARISON FOR CS

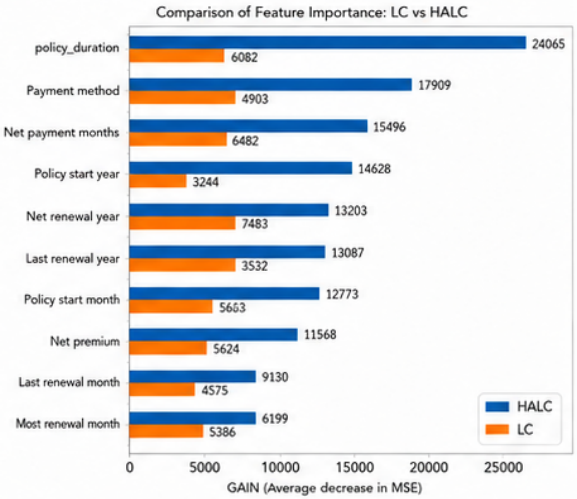


CLAIM LIKELIHOOD (CS) ROC-AUC (Validation Set) **0.8323**

Higher score indicates better model performance.

4 EXPLAINABLE RISK DRIVERS

Clear drivers help us price smarter and manage risk.



MODEL PERFORMANCE SUMMARY (VALIDATION SET)

RMSE (HALC Model)	RMSE (LC Model)
999	502

Lower RMSE indicates better accuracy in loss prediction.

5 BUSINESS RECOMMENDATION

Price with precision
Use predicted LC and HALC to align premiums with each policyholder's expected financial risk. Since HALC represents the estimated amount needed to break even, it can serve as a baseline for risk-adjusted pricing.

Protect low-risk customers
Avoid charging one flat premium across all policyholders. More accurate risk prediction helps prevent overpricing safer customers, reducing the chance that competitors attract them with cheaper rates.

Reward loyalty and stability
Because policy_duration is one of the strongest predictive features in the model, longer-tenured policyholders may signal lower or more stable risk. The company can offer renewal incentives, loyalty discounts, or retention programs to keep these stable customers.

Pricing accuracy is not a model metric— it's a profit lever.

DSO 530 Presentation
Section 16221
Group 3

Better risk prediction. Better pricing. Better results.